

Safety Case

Proposed deployment of a neural-network APU Health Monitor

Lab 5 · AI & Data Science with GenAI (Rail) — Deep Learning

Prepared by	
Date	
Notebook handed in	
System under assessment	Autoencoder-based APU Health Monitor (supplier proposal)
Evidence base	MetroPT-3, one unit, Feb–Jun 2020; two air-leak episodes; one held out for test

What this document is. A safety case is not a summary of what you did — it is an ARGUMENT that a senior engineer can disagree with, laid out so the disagreement lands in the right place. Write it for someone who understands trains, budgets and risk, and who has never heard of an autoencoder. Quote a number wherever it carries an argument. A safety case that is pasted code or pasted output scores no marks, and in the real world earns no signature.

1. The claim

Write the single sentence you are asking someone to sign. Be ruthless about scope — name the unit, the fault type, the evidence period, and the operating point. "The monitor detects APU failures" is not defensible on what you have; a narrower sentence that is actually true is worth far more. Then, underneath, state in one line what this system is NOT being claimed to do.

2. What the system is, in plain words

Explain to a non-specialist how the monitor works: what the network was trained to do, what it was and was not shown, what reconstruction error is, and how a number becomes an alarm. Four or five sentences. If you cannot explain it without the word "autoencoder", you do not yet understand it well enough to certify it.

3. Evidence FOR the claim

Fill in the table from Steps 6 and 7, then make the argument below it.

Measurement	Value	What it means operationally
ROC-AUC (smoothed score)		
Recall at the deployed threshold		
Precision at the deployed threshold		
False-alarm minutes in the test month		

Measurement	Value	What it means operationally
Missed failure minutes		

Now argue it. What does this evidence genuinely establish? Be specific about how the threshold was chosen and why that choice matters — a reviewer will ask whether you tuned it on the test set, and you need the answer ready.

4. Evidence AGAINST the claim

This section is the point of the document. A reviewer who finds a weakness you did not list stops trusting the whole case, including the parts that were right. Three specific findings from the lab belong here. A case that omits any of them cannot pass.

4.1 The no-learning baselines

Fill in the table from Step 8.

Detector (identical treatment: same smoothing, same threshold rule)	ROC-AUC	Recall	Precision	Missed	Learning required?
Deep autoencoder					Yes — 105,471 min
Control chart (any sensor beyond k SD)					None
DV_pressure on its own (Lab 2's sensor)					None

State plainly where the deep model finishes in that ranking. Then deal with the supplier's rebuttal — that the baselines only look good because you smoothed them too — using the raw-versus-smoothed figures from your Step 8 Your turn.

The Missed column — out of 390 failure minutes — is the one a safety reviewer reads first. Say what it shows. Then add the caveat that the winning detector is a rule aimed at the one fault type in this data, and say what that does and does not prove about the supplier's product.

4.2 The seed lottery

Fill in the spread you measured in Step 9.

Across five identical training runs	Lowest	Highest
ROC-AUC		
Recall		
False alarms		

Explain what this does to every number in section 3. Cover three things: that one of the five seeds produced a monitor which caught nothing at all; that the seed with the LOWEST training loss was among the worst performers on the actual job; and what both facts mean for change control — specifically, what has to happen if the supplier retrains the model after delivery.

4.3 The claim that has not been tested at all

The supplier's headline was that the monitor flags abnormal behaviour including fault types never seen before. From Challenge Part 2: describe the experiment that would test that claim, and state whether the evidence base in front of you can run it. Be precise about WHY not — the answer is structural, not a matter of sample size.

5. What it costs to be wrong — and who decided what "wrong" means

From Challenge Part 1. Two things in this section moved the answer, and neither of them is in the data.

5.1 The thresholds you happened to try

Threshold (percentile of training score)	False alarms	Missed	Total cost
90th			
95th			
97th			
98th			
99th			
99.9th			

Is that cost curve well behaved? Say what happens to the conclusion if the 98th percentile is left out of the grid — and what that tells you about any result that depends on a search grid somebody chose.

5.2 What is a false alarm: a minute, or a call-out?

Detector (all treated identically)	False-alarm minutes	False-alarm call-outs	Total cost
Deep autoencoder			
Control chart			
DV_pressure alone			

The two counts differ enormously for the same alarms — explain WHY (what does the 15-minute smoothing do to the shape of an alarm?), and say which unit reflects what the depot actually spends. Then report what Challenge

1(c) showed: the comparison that appeared to favour the deep model, why it was not a fair one, and what the ranking becomes once every detector is treated identically. Finally, the question that matters for this document: who chose the unit and the smoothing, and was either choice recorded before the numbers were produced?

6. Assumptions and limitations

What has to be true for your claim in section 1 to hold? Work through the table — add rows if you find more. For each, say what would happen if the assumption failed in service.

Assumption / limitation	What happens if it does not hold
The evidence is one unit, one fault type, two episodes	
The training period contains no undetected faults (we assumed those minutes were healthy)	
The record has gaps — the model has only seen running hours	
The test period differs from the training months in ways the data cannot attribute to a cause	
The 15-minute smoothing delays every alarm	

7. Residual risk

If this system is deployed and believed, what can still go wrong? Cover both directions of error and be concrete about consequences — an alarm that does not come, and an alarm that comes too often. Then answer the question that decides whether this document is worth anything: WHO carries each of these risks? Name the role, not the department.

Now the uncomfortable one. A monitor with low precision does not just waste time — it trains people to ignore it. Explain in two or three sentences how a health monitor can make a railway less safe than having no monitor at all, and say whether your proposed operating point runs that risk.

8. Recommendation

Choose one, and defend it. This is what you are putting your name to.

	Option	What it commits the depot to
☐	Deploy as a safety function	The depot may rely on it to decide what NOT to inspect
☐	Deploy as an advisory aid only	Alarms are investigated by a person; silence means nothing
☐	Trial under specified conditions	Runs alongside existing alarms, collecting evidence, deciding nothing
☐	Decline on current evidence	Not adopted; state what would change the answer

Your reasoning — one short paragraph. Your reasoning has to survive all three parts of Challenge 1 — the threshold grid, the cost unit, and the smoothing. Note how often this analysis appeared to favour the model only because of a choice the analyst made. A recommendation that does not account for that is not reading the evidence. Then, separately and specifically: what evidence would change your mind? A recommendation with no stated route to being reversed is a prejudice, not an engineering judgement.

Finally: if you recommend a trial or advisory use, list the conditions attached — what must be monitored, what must be recorded, what must trigger a re-assessment, and who reviews it.

9. Sign-off

Role	Name	Signature	Date
Prepared by (analyst)			
Reviewed by			
Accepted by (risk owner)			

By signing, the risk owner accepts the residual risk described in section 7 on the evidence described in sections 3 to 5.